

Astroinformatics 1: Graded Practice 1

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I. DOWNLOADING TESS DATA

The first step for this practice was using the "tesscurlsector73lc.sh" shell script to download 20 light curves in FITS format. Doing so was fairly straightforward, except I needed to look up how a shell script can be stopped mid process in order to avoid downloading too many files (press Ctrl+c on the terminal). Following this, we used TOPCAT to open the files and export them into CSV format. However, it should be noted that this process had to be done file by file as, for some reason, although TOPCAT apparently has the option to save multiple tables into a single file, it cannot save multiple tables into a file for each of them at the same time.

II. NAME MANAGING SHELL SCRIPTS

Now that we have obtained the 20 files in CSV format, the next step was to write a shell script to output a text file that contained all the names of the CSV files with the data of the light curves. The structure of the shell script (see figure 1) had three main steps: first, it created an output file and associated it with a variable. Then, it used the command list (ls) in combination with the wildcard to generate a list of all the files that ended with ".csv". Finally, it saved that list into the variable associated with the output file.

```
GNU nano 8.7.1
output_file="csv_files_names.txt"
ls *.csv > "$output_file"
```

Figure 1. Shell Script to read CSV files names.

Following this idea, we then wrote a second shell script (see figure 2) to read the previous output file, and generate a number of text files, each containing five names of the CSV files. The structure of this new shell script had a similar beginning, defining a variable for our now input file that we would read, and the number of names that each new file would have. Then there was the tricky part with the Split command, as it can

divide a file by many different parameters. In this case, the command received as input the variable with the number of names per file (5), the type of the generated files (.txt), the variable associated with our input file, and then the general name for the new files ("parte"). Now, we also used the command list to check that the expected number of new files was created (see figures 3, 4, 5 and 6).

```
GNU nano 8.7.1
input_file="csv_files_names.txt"
number_names=5

split -l $number_names -d --additional-suffix=.txt "$input_file" parte-
ls parte_*.txt
```

Figure 2. Shell Script to split the TXT file of names.

```
< > parte_00.txt parte_01.txt parte_02.txt parte_03.txt
home > jdgw > Documents > Tareas_ID > Astrofagasta > Magister > Materias > 1er Semestre > A
1 tess2023341045131-s0073-0000000001750268-0268-s_lc.csv
2 tess2023341045131-s0073-0000000001754086-0268-s_lc.csv
3 tess2023341045131-s0073-0000000001827744-0268-s_lc.csv
4 tess2023341045131-s0073-0000000001820070-0268-s_lc.csv
5 tess2023341045131-s0073-0000000001840666-0268-s_lc.csv
```

Figure 3. Text file with the first 5 names of the CSV files.

```
< > parte_00.txt parte_01.txt parte_02.txt parte_03.txt
home > jdgw > Documents > Tareas_ID > Astrofagasta > Magister > Materias > 1er Semestre > A
1 tess2023341045131-s0073-0000000001942416-0268-s_lc.csv
2 tess2023341045131-s0073-0000000001942969-0268-s_lc.csv
3 tess2023341045131-s0073-0000000001947463-0268-s_lc.csv
4 tess2023341045131-s0073-0000000001950736-0268-s_lc.csv
5 tess2023341045131-s0073-0000000001950093-0268-s_lc.csv
```

Figure 4. Text file with the second batch of 5 names of the CSV files.

```
< > parte_00.txt parte_01.txt parte_02.txt parte_03.txt
home > jdgw > Documents > Tareas_ID > Astrofagasta > Magister > Materias > 1er Semestre > A
1 tess2023341045131-s0073-0000000002060684-0268-s_lc.csv
2 tess2023341045131-s0073-0000000002080765-0268-s_lc.csv
3 tess2023341045131-s0073-000000000214191-0268-s_lc.csv
4 tess2023341045131-s0073-0000000002102329-0268-s_lc.csv
5 tess2023341045131-s0073-0000000002104696-0268-s_lc.csv
```

Figure 5. Text file with the third batch of 5 names of the CSV files.

```
< > parte_00.txt parte_01.txt parte_02.txt parte_03.txt
home > jdgw > Documents > Tareas_ID > Astrofagasta > Magister > Materias > 1er Semestre > A
1 tess2023341045131-s0073-0000000002105589-0268-s_lc.csv
2 tess2023341045131-s0073-0000000002143575-0268-s_lc.csv
3 tess2023341045131-s0073-0000000002140970-0268-s_lc.csv
4 tess2023341045131-s0073-0000000002152411-0268-s_lc.csv
5 tess2023341045131-s0073-0000000002234692-0268-s_lc.csv
```

Figure 6. Text file with the fourth batch of 5 names of the CSV files.

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III. LIGHT CURVES PLOTS

Afterwards, the last step was to plot the light curves using the TOPCAT software. As part of this process, I tried to find information related to the data that each column had in the CSV file; my first idea was to check the header of the FITS files, but sadly, those contained information related to the observation itself and not how the data was structured in the file. Therefore, I checked the MAST documentation, specifically the section related to the TESS light curves (see https://spacetelescope.github.io/mastr_notebooks/notebooks/TESS/beginner_how_to_use_lc/beginner_how_to_use_lc.html), which described the columns for our CSV files, the most important ones for plotting the light curves were the following:

1. Time
2. SAP Flux: Simple Aperture Photometry flux, the flux after summing the calibrated pixels within the TESS optimal photometric aperture.
3. PDCSAP Flux: Pre-search Data Conditioned Simple Aperture Photometry, the SAP flux values nominally corrected for instrumental variations. Thus, these fluxes are the mission's best estimate of the intrinsic variability of the target.

For plotting the light curves, we will always put Time on the X axis, and in this case, we will put either kind of flux according to each file:

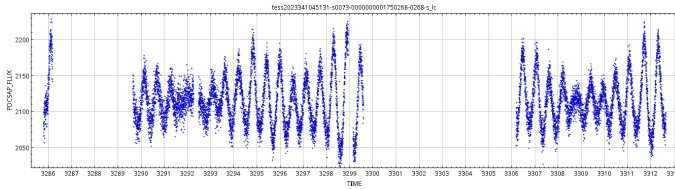


Figure 7. Light curve of file 1. Notice in Y axis the PDCSAP Flux, so we can see easily the intrinsic variation in the luminosity of the source.

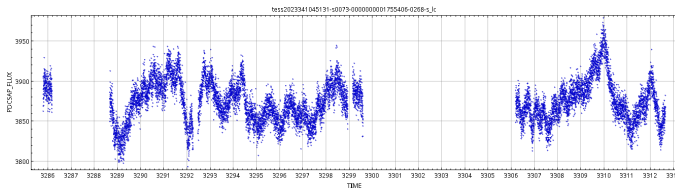


Figure 8. Light curve of file 2. Notice in Y axis the PDCSAP Flux, so we can see easily the intrinsic variation in the luminosity of the source.

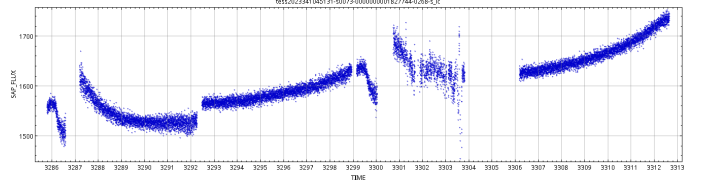


Figure 9. Light curve of file 3. Notice in Y axis the SAP Flux, so we can see the variation in the luminosity due to the instrument, as the PDCSAP Flux did not show any kind of signal.

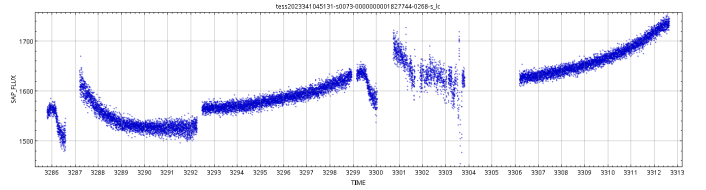


Figure 10. Light curve of file 4. Notice in Y axis the SAP Flux, so we can see the variation in the luminosity due to the instrument, as the PDCSAP Flux did not show any kind of signal.

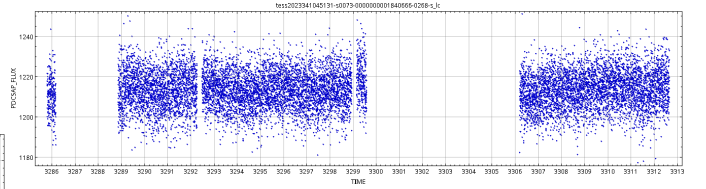


Figure 11. Light curve of file 5. Notice in Y axis the PDCSAP Flux, so we can see noise in the luminosity without any kind of signal. Actually the light curve of the SAP Flux was similar to the ones of Figure 9 and 10.

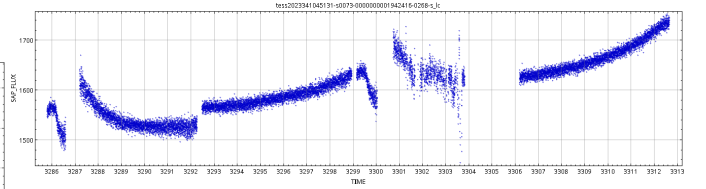


Figure 12. Light curve of file 6. Notice in Y axis the SAP Flux. As can be easily seen most sources did not have a signal that could be easily read on the PDCSAP Flux light curves, as a matter of fact, light curves from files 7 to 15 exhibited this same behaviour.

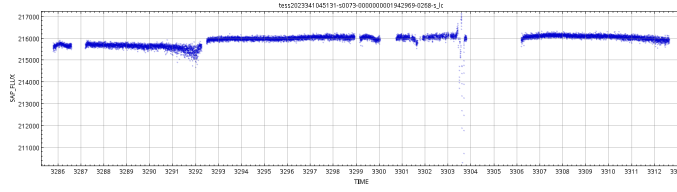


Figure 13. Light curve of file 16. Notice in Y axis the SAP Flux, so we can see the variation in the luminosity due to the instrument which was lower than in previous figures. Still, PDCSAP Flux did not show any kind of signal.

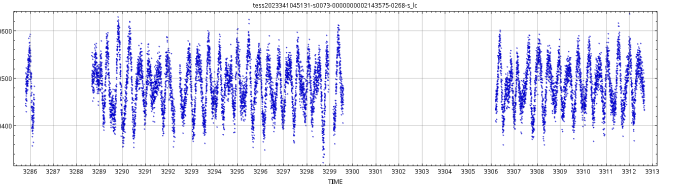


Figure 15. Light curve of file 18. Notice in Y axis the PDCSAP Flux, so we can see easily the variation in the intrinsic luminosity of the source.

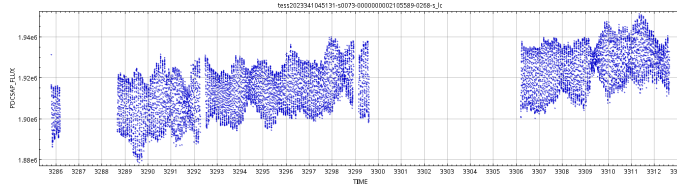


Figure 14. Light curve of file 17. Notice in Y axis the PDCSAP Flux, so we can see this weird patron on the variation of the luminosity, which in theory is not a product of the instrument as it persisted after the corrections for instrumental noise, so it must be a consequence of the nature of the source.

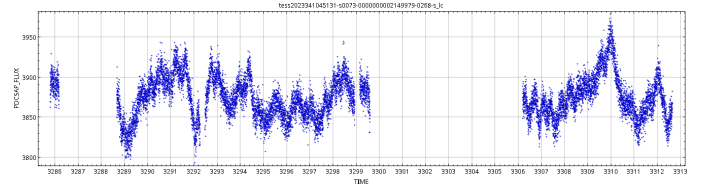


Figure 16. Light curve of file 19. Notice in Y axis the PDCSAP Flux, so we can see easily the variation in the intrinsic luminosity of the source.

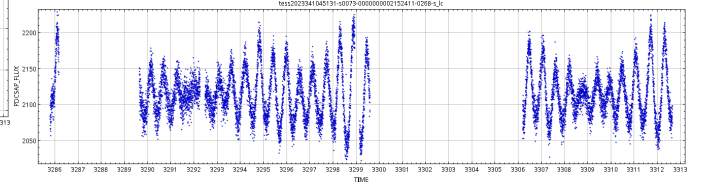


Figure 17. Light curve of file 20. Notice in Y axis the PDCSAP Flux, so we can see easily the variation in the intrinsic luminosity of the source.